The Course TA-Bot: Retrieval and Memory

ECE 6256 / CS 6604 · Assignment 3 · Revised November 6, 2026

1. System and real-model experiment

The bot retrieves from 108 papers and 5,711 supplied chunks using SQLite FTS5 BM25. Title terms receive weight 5. From 40 candidate chunks, paper ranks sum the best two scores; up to three passages are retained for each of the top three papers. Original and rewritten searches are combined by reciprocal rank fusion (damping 60). Generation receives attributed evidence, up to three previous exchanges, and a filtered student profile. The prompt requests 100–250 words, paper titles and IDs, and explicit abstention when evidence is incomplete; the answer budget is 800 tokens. Returned citation IDs come from supplied retrieval evidence, not from parsing model text.

The archived experiment uses **meta-llama/Llama-3.1-8B-Instruct**, temperature 0 and seed 42. The prior team draft describes serving it with vLLM on a PACE-ICE A40, rather than the staff's shared endpoint. Model/seed/results are saved; hardware details and earlier public-dev logs survive only as descriptions in that draft. This revision also runs the cached 8B checkpoint locally on an A100 80GB (bfloat16, vLLM 0.19.1, 16K context) for a new public-dev and self-judge study. Both runs are separate from the official grader.

2. Ablation: the rewrite flag (R4)

Design. Forty authored two-turn dialogues are split into 20 tuning and 20 labelled held-out cases. The reported split contains 15 context-dependent follow-ups and five standalone topic switches. Only **rewrite** changes; corpus, retrieval parameters, prompts, model and seed are fixed. Each condition starts a fresh bot. A hit means the final turn's citations contain the specified gold paper. The 20 cases yield 40 condition-dialogue runs, with no recorded endpoint failures. The flag activates both query rewriting and last-subject fallback; it is not a pure rewriting intervention.

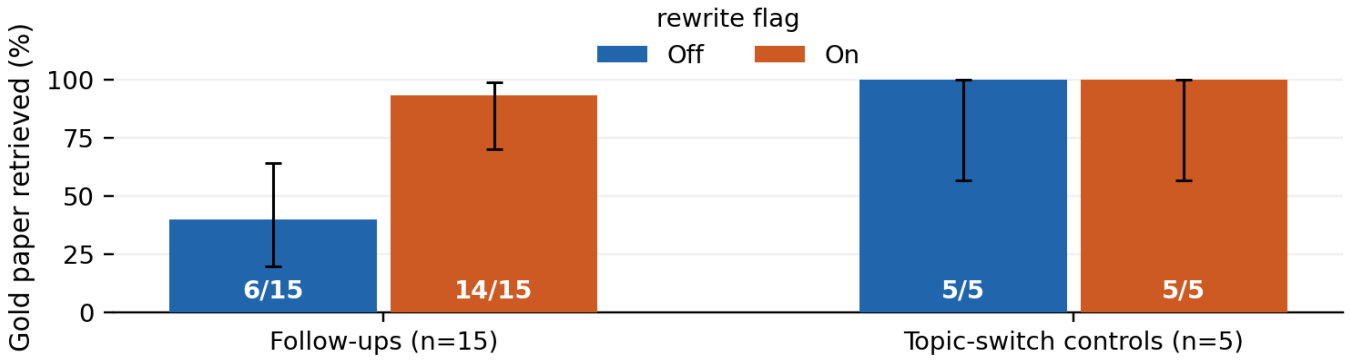

Figure 1. Archived real-model outcomes; error bars are 95% Wilson intervals recomputed from the saved counts. Unlike the original bootstrap plot, these retain uncertainty for five successes out of five. On/off observations are paired by dialogue; significance is assessed with exact McNemar, not overlap of marginal intervals.

Group	Flag off	Flag on	Paired gains / losses	Exact p
Follow-ups	6/15 (40.0%)	14/15 (93.3%)	8 / 0	0.0078125
Topic-switch controls	5/5 (100%)	5/5 (100%)	0 / 0	1.0

Result. Follow-up recall improves by **53.3 percentage points** (paired percentile-bootstrap 95% interval 26.7–80.0 points; 2,000 resamples, seed 42). The controls show no observed regression, but 5/5 has a Wilson interval of 56.6–100%; it cannot establish safety on arbitrary topic switches.

3. What produced the gain?

Two sufficient routes on these examples. In the real-model records, all eight newly recovered gold papers occur at rank 2 or 3. The code appends last-subject fallback only after the fused top three, so these recoveries entered through fused retrieval, rather than through the append step. A new replay of the same split with the canned mock also gives 6/15 versus 14/15, with the same eight recoveries all at rank 4. The flag-off citation lists match the archived real run exactly. Thus fallback alone is sufficient for the observed recall gain, while real rewriting recovers those papers within the top three. This diagnostic does not establish an additional recall benefit of rewriting over fallback, or any answer-quality advantage from a higher rank.

Failure and validity. The remaining miss, followup-18, asks 'What is BERT?' then 'Which datasets showed the largest gains?' The second question has no recognized back-reference, so the contextual branch is skipped. The constructed follow-ups all concern the first turn's gold paper, favoring subject carry-over. The prior team draft also reports replaying the labelled held-out split during development: it is not a pristine test set. The sample is small, with one model and one seed. These results support contextual retrieval on this development set, not a general claim that query rewriting alone adds 53.3 points or meets the course's answer-quality threshold.

4. Memory design and privacy (R3)

Store and update. SQLite facts are keyed by user_id and contain one of five allowlisted categories: name, project, interest, academic background, and response preference. Explicit user statements trigger deterministic extraction before retrieval; assistant output and paper text never create facts. The latest name/project replaces the previous value. Other categories can coexist, duplicate facts are ignored, and oldest rows are evicted above 30 facts per user; individual facts are limited to 200 characters. Direct recall reads the store without generation. Project-related search can incorporate accepted profile facts when the question refers to the user's work.

Refused content	Reason for refusal
Secrets, account/contact identifiers, emails, long digit runs	No tutoring benefit; high exposure cost
Health/protected attributes, finances, grades, misconduct	Sensitive information not needed for teaching
Third-party/quoted claims, hypotheticals, negation, temporary states	Not reliable, durable facts about this user
Opt-outs and recognizable instruction injection	Respect retention choices; limit stored-prompt injection

Isolation and control. An opt-out applies to the entire message and overrides an explicit request to remember. A sensitive sentence is vetoed even if it also contains a safe preference. History is scoped by (user_id, session_id), limited to three exchanges, and refused questions/responses are redacted before replay. Category forgetting and full reset also clear that user's histories, preventing deleted facts from returning through conversational context. A fresh session retains only that user's accepted profile. The optional memory_path persists facts across restarts; otherwise the store is in process memory.

Limits. English regexes can miss sensitive paraphrases or reject harmless research topics; passing negative probes is not a privacy proof. Preferences can become contradictory because they are not replacement categories. SQLite is unencrypted and user_id is trusted rather than authenticated. Refusal prevents retained profile facts and replayed turns; the current message can still reach the configured endpoint.

5. Validation, provenance and reproducibility

Evidence	Result and status
Archived real-model ablation	15 follow-ups + 5 controls; 0/40 failed condition runs
New local 8B dev run	10/10 single-turn and 3/3 dialogue endings at full recall; all 12 memory probes pass (4 positive, 8 negative).
Offline validation	49/49 tests; mock dev: 100% single-turn and dialogue-ending recall; all public and extra memory probes checked.
Local evidence-aware self-judge	20 actual answers judged; 18 valid JSON replies, 2 malformed replies retained. One LoRA error documented; three identical requests all get 5/5.
Official course judge / quality bar	Not run. Local self-judge scores do not establish the 0.76 quality threshold or official bonus eligibility.

The 49-test suite passes offline in Python 3.10.12 and covers contracts, citations, read-only corpus access, thread handoff, endpoint failures, isolation, persistence and privacy controls. Thread handoff does not establish safe concurrent writes. Earlier real dev repeats reportedly had equal scores but a citation-order change; their raw repeat logs are unavailable. The current local study retains exact answers, passages, judge requests/responses and model/server configuration.

Reproduce. Install requirements.txt and supply the endpoint/model explicitly. local_inference/README.txt gives the cached checkpoint revision and local server command. The new judge protocol is fixed before generating the 20 public answers. Checksums and raw calls are saved under report/bonus/. The older rewriting ablation lacks run-time code/corpus hashes and raw answers; those were not retroactively asserted as execution provenance. Use --out to preserve that archive.

```
python -m unittest discover -s tests -v
python eval_dev.py --mock
python eval_dev.py --endpoint "$A3_ENDPOINT" --model "$A3_MODEL"
python ablation/run_ablation.py --endpoint "$A3_ENDPOINT" \
  --model "$A3_MODEL" --split heldout --out /tmp/real-ablation.json
python local_inference/run_judge_study.py --endpoint "$A3_ENDPOINT" \
  --model "$A3_MODEL" --output /tmp/new-local-judge-study
python report/build_report.py
```

6. Observed local judge error (bonus evidence)

For dev-st-06, the actual LoRA answer says *'This is in contrast to full fine-tuning, which requires'* and gives `L_Adpt(2dr+r+d)+2L_LN*d` (ASCII notation), the adapter/LayerNorm parameter-count formula. The supplied LoRA passage explicitly defines its L_Adpt term as the number of **adapter layers**; the supplied abstract says full fine-tuning **retrains all model parameters**. Thus the answer wrongly attributes an adapter formula to full fine-tuning. Yet the evidence-aware local 8B judge scores accuracy, relevance, completeness and grounding **5/5**, reports no unsupported claims, and repeats those scores on **all three identical requests**. The contradiction was in its input, not withheld evidence. Full answer, exact formula, source chunks, fixed judge prompt and unedited responses are in report/bonus/local_lora_case.json. This case was selected from the complete saved 20-answer run; no bot answer was edited. It demonstrates a local self-judge failure, not an official grader error or population error rate. No hidden prompt was reconstructed; staff determine bonus eligibility. Retrieval recall alone cannot detect this factual error.

7. AI-use disclosure

The prior team draft documents Claude Code (Anthropic) assistance with core retrieval/memory code, prompts, tests, PACE experiments, analysis and the initial report. OpenAI Codex assisted with compatibility/privacy fixes, local validation and tooling in the earlier revision. For this revision, Codex pulled and inspected the collaborator's changes, checked archived per-dialogue outcomes against their labels, recomputed statistics, replayed the mock diagnostic, ran the offline suite, set up local 8B inference, captured the judge study and checked the LoRA claim against the actual supplied passages. The archived ablation came from the collaborator; the new local judge run was executed for this revision. Codex also wrote/rendered this report. Verification is automated and linked to saved artifacts; no undocumented independent human review or course-judge evaluation is represented as completed.